

InstructBLIP: 基于指令微调的通用视觉-语言模型

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<https://github.com/salesforce/LAVIS/tree/main/projects/instructblip>

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Abstract

大规模预训练与指令微调已成功打造出具备广泛能力的通用语言模型。然而，由于引入视觉输入后带来的丰富输入分布与任务多样性，构建通用视觉-语言模型面临更大挑战。尽管视觉-语言预训练已得到广泛研究，视觉-语言指令微调领域仍有待深入探索。本文基于预训练的 BLIP-2 模型，对视觉-语言指令微调进行了系统而全面的研究。我们收集了 26 个公开数据集，涵盖多种任务类型与能力，并将其转换为指令微调格式。此外，我们引入了一种指令感知的 Query Transformer，能够根据给定指令提取与之匹配的信息特征。在 13 个 held-in 数据集上训练后，InstructBLIP 在全部 13 个 held-out 数据集上均达到了最优的零样本性能，大幅超越 BLIP-2 及更大规模的 Flamingo 模型。在针对单个下游任务进行微调时，我们的模型同样达到了最优性能（例如，在含图像上下文的 ScienceQA 问题上准确率达 90.7%）。此外，我们通过定性分析展示了 InstructBLIP 相较于同期多模态模型的优势。所有 InstructBLIP 模型均已开源。

1 引言

人工智能（AI）研究的长期愿景是构建单一模型，使其能够解决用户指定的任意任务。在自然语言处理（NLP）领域，指令微调 [46, 7] 被证明是实现这一目标的有效途径。通过在大量自然语言指令描述的任务上微调大语言模型（LLM），指令微调使模型能够遵循任意指令。近年来，经过指令微调的 LLM 也被引入视觉-语言任务中。例如，BLIP-2 [20] 能够有效地使冻结的指令微调 LLM 理解视觉输入，并展示了在图像到文本生成中遵循指令的初步能力。

相比 NLP 任务，视觉-语言任务由于引入了来自不同领域的视觉输入，天然具有更大的多样性。这对一个需要泛化到多种视觉-语言任务（其中许多在训练中未曾见过）的统一模型提出了更高挑战。现有方法大致可分为两类：第一类是多任务学习 [6, 27]，将各种视觉-语言任务统一为相同的输入-输出格式。然而，我们通过实验发现，不含指令的多任务学习（表 4）对未见数据集和任务的泛化能力较差。第二类方法 [20, 4] 在预训练 LLM 基础上引入视觉组件，并使用图像描述数据训练视觉组件。然而，此类数据的多样性不足，难以支撑对超出视觉描述范围的视觉-语言任务的广泛泛化。

为解决上述挑战，本文提出 InstructBLIP —— 一个视觉-语言指令微调框架，使通用模型能够通过统一的自然语言接口解决多种视觉-语言任务。InstructBLIP 使用多样化的指令数据训练多模态 LLM。具体而言，我们基于预训练的 BLIP-2 模型进行初始化，该模型由图像编码器、LLM 以及连接二者的 Query Transformer（Q-Former）组成。在指令微调阶段，我们仅微调 Q-Former，同时保持图像编码器和 LLM 冻结。本文的主要贡献如下：

- 我们对视觉-语言指令微调进行了全面系统的研究。我们将 26 个数据集转换为指令微调格式，并归纳为 11 个任务类别。其中 13 个 held-in 数据集用于指令微调，13 个 held-out 数

*Work done during internship at Salesforce.



Figure 1: 我们的 InstructBLIP Vicuna 模型生成的若干定性示例，展示了其多种能力，包括复杂视觉场景理解与推理、基于知识的图像描述、多轮视觉对话等。

数据集用于零样本评估。此外，我们还完整保留了四个任务类别用于任务级零样本评估。大量定量与定性实验结果验证了 InstructBLIP 在视觉-语言零样本泛化上的有效性。

- 我们提出了指令感知的视觉特征提取机制——一种根据给定指令灵活提取信息特征的新机制。具体来说，文本指令不仅输入到冻结的 LLM 中，还同时输入到 Q-Former，使其能够从冻结的图像编码器中提取与指令相关的视觉特征。此外，我们还提出了平衡采样策略，以同步各数据集的学习进度。
- 我们评估并开源了一系列基于两类 LLM 的 InstructBLIP 模型：1) FlanT5 [7]，一种基于 T5 [34] 微调的编码器-解码器 LLM；2) Vicuna [2]，一种基于 LLaMA [41] 微调的仅解码器 LLM。InstructBLIP 模型在多种视觉-语言任务上实现了最优的零样本性能。此外，将 InstructBLIP 用作模型初始化权重时，在各个下游任务的微调性能上同样达到最优水平。

2 视觉-语言指令微调

InstructBLIP 旨在解决视觉-语言指令微调中的独特挑战，并对模型在未见数据和任务上的泛化能力提升进行系统研究。本节首先介绍指令微调数据的构建方式，随后阐述训练与评估协议。接着，分别从模型和数据两个角度介绍两种提升指令微调性能的技术。最后，给出实现细节。

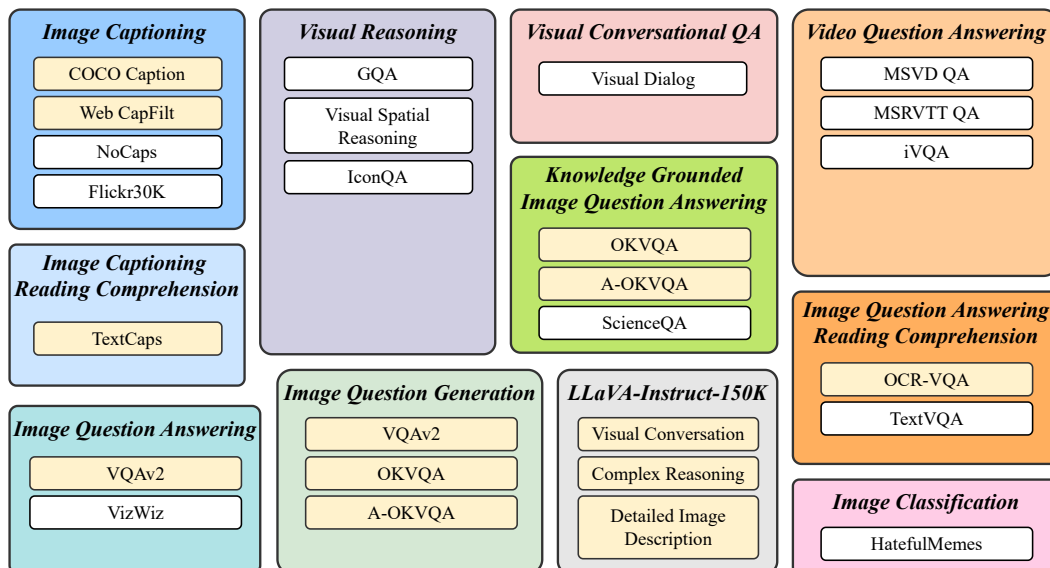


Figure 2: 用于视觉-语言指令微调的任务及其对应数据集。黄色表示 held-in 数据集，白色表示 held-out 数据集。

2.1 任务与数据集

为在保证可获取性的前提下确保指令微调数据的多样性，我们收集了一批公开的视觉-语言数据集，并将其转换为指令微调格式。如图 2 所示，最终收集涵盖 11 个任务类别和 26 个数据集，包括图像描述 [23, 3, 51]、含阅读理解的图像描述 [38]、视觉推理 [16, 24, 29]、图像问答 [11, 12]、知识增强的图像问答 [30, 36, 28]、含阅读理解的图像问答 [31, 39]、图像问题生成（改编自 QA 数据集）、视频问答 [47, 49]、视觉对话问答 [8]、图像分类 [18] 以及 LLaVA-Instruct-150K [25]。各数据集的详细描述和统计信息见附录 C。

针对每个任务，我们精心设计了 10 至 15 个不同的自然语言指令模板。这些模板是构建指令微调数据的基础，用于阐明任务目标。对于原本倾向于短回答的公开数据集，我们在部分对应指令模板中加入简短、简要等词汇，以降低模型始终生成短输出的过拟合风险。对于 LLaVA-Instruct-150K 数据集，由于其本身已采用指令格式，故不额外添加指令模板。完整的指令模板列表见附录 D。

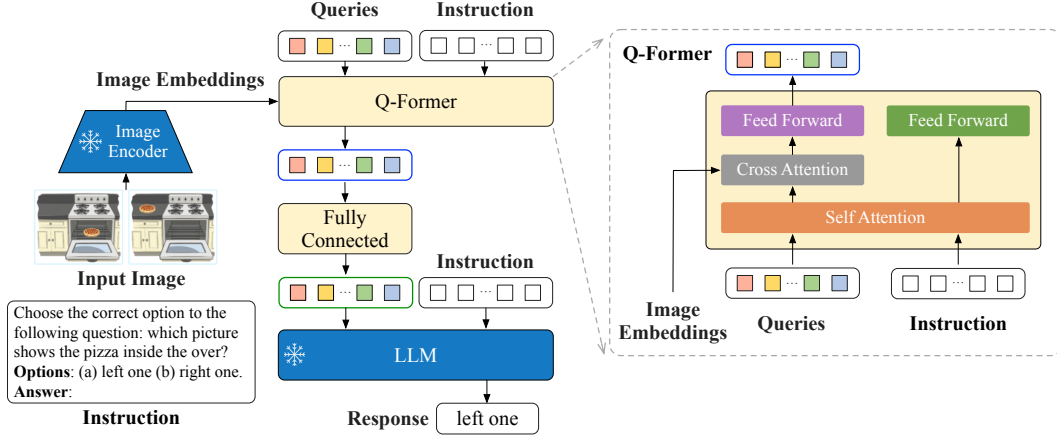


Figure 3: Model architecture of InstructBLIP. The Q-Former extracts instruction-aware visual features from the output embeddings of the frozen image encoder, and feeds the visual features as soft prompt input to the frozen LLM. We instruction-tune the model with the language modeling loss to generate the response.

2.2 训练与评估协议

为保证训练和零样本评估有充足的数据与任务，我们将 26 个数据集划分为 13 个 held-in 数据集和 13 个 held-out 数据集，在图 2 中分别以黄色和白色标注。我们使用 held-in 数据集的训练集进行指令微调，使用其验证集或测试集进行 held-in 评估。

对于 held-out 评估，我们的目标是理解指令微调如何提升模型在未见数据上的零样本性能。我们定义了两类 held-out 数据：1) 训练阶段模型未见过的数据集，但其对应任务存在于 held-in 集合中；2) 训练阶段完全未见过的数据集及其关联任务。由于 held-in 与 held-out 数据集之间存在数据分布偏移，第一类 held-out 评估本身具有一定难度。对于第二类，我们完整保留了若干任务类别，包括视觉推理、视频问答、视觉对话问答和图像分类。

为避免数据污染，我们仔细筛选数据集，确保不同数据集的评估数据不出现在 held-in 训练集中。在指令微调阶段，我们混合所有 held-in 训练集，并对每个数据集均匀采样指令模板，以标准语言建模损失直接训练模型根据指令生成回答。此外，对于涉及场景文本的数据集，我们在指令中额外加入 OCR token 作为补充信息。

2.3 指令感知的视觉特征提取

现有零样本图像到文本生成方法（包括 BLIP-2）在提取视觉特征时采用与指令无关的方式，导致无论任务如何，输入 LLM 的视觉表示始终是静态的。相比之下，具备指令感知能力的视觉模型能够根据任务指令自适应地生成最有利于当前任务的视觉表示。当同一输入图像对应的任务指令差异显著时，这一优势尤为明显。

InstructBLIP 的架构如图 3 所示。与 BLIP-2 [20] 类似，InstructBLIP 使用 Query Transformer（即 Q-Former）从冻结的图像编码器中提取视觉特征。Q-Former 的输入包含一组 K 个可学习的查询嵌入，通过交叉注意力与图像编码器的输出进行交互。Q-Former 的输出为 K 个编码后的视觉向量（每个查询嵌入对应一个），经线性投影后输入到冻结的 LLM 中。与 BLIP-2 相同，Q-Former 在指令微调前先用图像-描述数据进行两阶段预训练：第一阶段与冻结的图像编码器联合预训练，用于视觉-语言表示学习；第二阶段将 Q-Former 的输出适配为冻结 LLM 的软视觉提示，用于文本生成。预训练完成后，通过指令微调对 Q-Former 进行微调，此时 LLM 接收来自 Q-Former 的视觉编码和任务指令作为输入。

在 BLIP-2 基础上，InstructBLIP 提出了指令感知的 Q-Former 模块，将指令文本 token 作为额外输入。指令通过 Q-Former 的自注意力层与查询嵌入进行交互，引导其提取与任务相关的图像特征，从而使 LLM 获得更有利于遵循指令的视觉信息。实验结果（表 2）表明，指令感知的视觉特征提取在 held-in 和 held-out 评估上均带来了显著的性能提升。

2.4 训练数据集平衡

由于训练数据集数量众多且各数据集规模差异显著，均匀混合可能导致模型在小数据集上过拟合、在大数据集上欠拟合。为缓解这一问题，我们提出按数据集规模的平方根比例进行采样。具体而言，给定 D 个规模为 $\{S_1, S_2, \dots, S_D\}$ 的数据集，训练时从数据集 d 中选取样本的概率为 $p_d = \frac{\sqrt{S_d}}{\sum_{i=1}^D \sqrt{S_i}}$ 。在此公式基础上，我们对部分数据集的权重进行手动调整以改善优化效果。这是因为不同数据集和任务之间存在固有差异，即便规模相近，所需的训练强度也可能不同。具体而言，我们降低了 A-OKVQA（多选题形式）的权重，并提高了 OKVQA（开放式文本生成）的权重。表 2 的结果表明，平衡数据集采样策略在 held-in 评估和 held-out 泛化上均提升了整体性能。

2.5 推理方法

在推理阶段，我们针对不同数据集采用两种略有差异的生成方式进行评估。对于大多数数据集（如图像描述和开放式 VQA），直接提示经指令微调的模型生成回答，再与标准答案对比计算指标。对于分类和多选 VQA 任务，我们参考已有工作 [46, 22, 21] 采用词汇排序方法：仍提示模型生成答案，但将词汇范围限定为候选列表，计算每个候选项的对数似然值，取最高者作为最终预测。该排序方法应用于 ScienceQA、IconQA、A-OKVQA（多选）、HatefulMememes、Visual Dialog、MSVD 和 MSRVT 数据集。此外，对于二分类任务，我们将正负标签扩展为略微宽泛的词汇集以利用自然文本中的词频（例如，正类使用 *yes* 和 *true*，负类使用 *no* 和 *false*）。

对于视频问答任务，每个视频均匀采样四帧，每帧分别经图像编码器和 Q-Former 独立处理，提取的视觉特征拼接后输入 LLM。

2.6 实现细节

模型架构。 得益于 BLIP-2 模块化架构设计的灵活性，我们可以快速将模型适配到多种 LLM 上。实验中，我们采用四种 BLIP-2 变体，均使用相同的图像编码器（ViT-g/14 [10]），但冻结 LLM 不同，包括 FlanT5-XL (3B)、FlanT5-XXL (11B)、Vicuna-7B 和 Vicuna-13B。FlanT5 [7] 是基于编码器-解码器 Transformer T5 [34] 微调的指令微调模型。Vicuna [2] 则是近期发布的、基于 LLaMA [41] 微调的仅解码器 Transformer 指令微调模型。在视觉-语言指令微调阶段，我们从预训练的 BLIP-2 检查点初始化模型，仅微调 Q-Former 的参数，同时保持图像编码器和 LLM 均冻结。由于原始 BLIP-2 模型不含 Vicuna 对应的检查点，我们按照与 BLIP-2 相同的流程对 Vicuna 进行预训练。

训练与超参数。 我们使用 LAVIS 库 [19] 进行实现、训练和评估。所有模型最多指令微调 60K 步，每 3K 步验证一次性能。每个模型选取单一最优检查点用于所有数据集的评估。3B、7B 和 11/13B 模型的批次大小分别为 192、128 和 64。优化器使用 AdamW [26]，参数设置为 $\beta_1 = 0.9$ 、 $\beta_2 = 0.999$ 、权重衰减 0.05。学习率在前 1,000 步进行线性预热，从 10^{-8} 增至 10^{-5} ，随后按余弦衰减至最小学习率 0。所有模型在 16 块 Nvidia A100 (40G) GPU 上训练完成，耗时约 1.5 天。

3 实验结果与分析

3.1 零样本评估

我们首先在 13 个 held-out 数据集上评估 InstructBLIP 模型，所用指令见附录 E。我们将 InstructBLIP 与此前最优模型 BLIP-2 和 Flamingo 进行对比。如表 1 所示，我们在所有数据集上均取得了新的零样本最优结果。InstructBLIP 在所有 LLM 配置下均以显著优势超越其基础模型 BLIP-2，充分验证了视觉-语言指令微调的有效性。例如，InstructBLIP FlanT5_{XL} 相比 BLIP-2 FlanT5_{XL} 平均相对提升达 15.0%。此外，指令微调显著提升了模型对未见任务类别（如视频问答）的零样本泛化能力。尽管从未在时序视频数据上训练，InstructBLIP 在 MSRVT-QA 上相比此前最优结果实现了高达 47.1% 的相对提升。最后，参数量仅 4B 的最小版本 InstructBLIP FlanT5_{XL} 在 6 个共同评估数据集上均超越 Flamingo-80B，平均相对提升 24.8%。

对于 Visual Dialog 数据集，我们选择报告平均倒数排名（MRR）而非归一化折扣累积增益（NDCG）指标。这是因为 NDCG 倾向于奖励模糊和不确定的答案，而 MRR 更偏向确定性回答 [32]，因此 MRR 更符合零样本评估场景的特点。

	NoCaps	Flickr 30K	GQA	VSR	IconQA	TextVQA	Visdial	HM	VizWiz	SciQA image	MSVD QA	MSRVT QA	iVQA
Flamingo-3B [4]	-	60.6	-	-	-	30.1	-	53.7	28.9	-	27.5	11.0	32.7
Flamingo-9B [4]	-	61.5	-	-	-	31.8	-	57.0	28.8	-	30.2	13.7	35.2
Flamingo-80B [4]	-	67.2	-	-	-	35.0	-	46.4	31.6	-	35.6	17.4	40.7
BLIP-2 (FlanT5 _{XL}) [20]	104.5	76.1	44.0	60.5	45.5	43.1	45.7	53.0	29.8	54.9	33.7	16.2	40.4
BLIP-2 (FlanT5 _{XXL}) [20]	98.4	73.7	44.6	68.2	45.4	44.1	46.9	52.0	29.4	64.5	34.4	17.4	45.8
BLIP-2 (Vicuna-7B)	107.5	74.9	38.6	50.0	39.7	40.1	44.9	50.6	25.3	53.8	18.3	9.2	27.5
BLIP-2 (Vicuna-13B)	103.9	71.6	41.0	50.9	40.6	42.5	45.1	53.7	19.6	61.0	20.3	10.3	23.5
InstructBLIP (FlanT5 _{XL})	119.9	84.5	48.4	64.8	50.0	46.6	46.6	56.6	32.7	70.4	43.4	25.0	53.1
InstructBLIP (FlanT5 _{XXL})	120.0	83.5	47.9	65.6	51.2	46.6	48.5	54.1	30.9	70.6	44.3	25.6	53.8
InstructBLIP (Vicuna-7B)	123.1	82.4	49.2	54.3	43.1	50.1	45.2	59.6	34.5	60.5	41.8	22.1	52.2
InstructBLIP (Vicuna-13B)	121.9	82.8	49.5	52.1	44.8	50.7	45.4	57.5	33.4	63.1	41.2	24.8	51.0

Table 1: held-out 数据集上的零样本结果。其中 Visdial、HM 和 SciQA 分别代表 Visual Dialog、HatefulMemes 和 ScienceQA 数据集。对于 ScienceQA，仅评估含图像上下文的子集。参照已有工作 [4, 49, 32]，NoCaps 和 Flickr30K 报告 CIDEr 分数 [42]，iVQA 报告准确率，HatefulMemes 报告 AUC 分数，Visual Dialog 报告平均倒数排名（MRR）。其余数据集均报告 top-1 准确率（%）。

Model	Held-in Avg.	GQA	ScienceQA (image-context)	IconQA	VizWiz	iVQA
InstructBLIP (FlanT5 _{XL})	94.1	48.4	70.4	50.0	32.7	53.1
w/o Instruction-aware Visual Features	89.8	45.9 (↓2.5)	63.4 (↓7.0)	45.8 (↓4.2)	25.1 (↓7.6)	47.5 (↓5.6)
w/o Data Balancing	92.6	46.8 (↓1.6)	66.0 (↓4.4)	49.9 (↓0.1)	31.8 (↓0.9)	51.1 (↓0.0)
InstructBLIP (Vicuna-7B)	100.8	49.2	60.5	43.1	34.5	52.2
w/o Instruction-aware Visual Features	98.9	48.2 (↓1.0)	55.2 (↓5.3)	41.2 (↓1.9)	32.4 (↓2.1)	36.8 (↓15.4)
w/o Data Balancing	98.8	47.8 (↓1.4)	59.4 (↓1.1)	43.5 (↓0.4)	32.3 (↓2.2)	50.3 (↓1.9)

Table 2: 分别移除指令感知视觉特征（第 2.3 节）和平衡数据集采样策略（第 2.4 节）的消融实验结果。held-in 评估取 COCO Caption、OKVQA、A-OKVQA 和 TextCaps 四个数据集的平均分；held-out 评估展示来自不同任务的五个数据集结果。

3.2 指令微调技术消融实验

为研究指令感知视觉特征提取（第 2.3 节）和平衡数据集采样策略（第 2.4 节）的影响，我们在指令微调过程中开展了消融实验。如表 2 所示，移除视觉特征中的指令感知能力会导致所有数据集上的性能显著下降。性能下降在涉及空间视觉推理（如 ScienceQA）或时序视觉推理（如 iVQA）的数据集上更为明显——在这些任务中，Q-Former 中的指令输入能够引导视觉特征关注图像中的关键区域。移除数据平衡策略则会导致训练不稳定且不均衡，不同数据集在差异悬殊的训练步数时才各自达到最优性能，多数据集之间学习进度缺乏同步，进而损害整体性能。

3.3 定性评估

除在公开基准上的系统性评估外，我们进一步使用更多样化的图像和指令对 InstructBLIP 进行定性分析。如图 1 所示，InstructBLIP 展现出强大的复杂视觉推理能力：例如，它能从视觉场景中合理推断可能发生了什么，并根据棕榈树等视觉证据推断场景地点，进而判断灾害类型。此外，InstructBLIP 能够将视觉输入与内嵌的文本知识相结合，生成富有信息量的回答，例如介绍一幅名画。在对整体氛围的描述中，InstructBLIP 还展现出理解视觉图像隐喻含义的能力。最后，我们展示了 InstructBLIP 能够进行多轮对话，在生成新回答时有效地考虑对话历史。

在附录 B 中，我们将 InstructBLIP 与同期多模态模型（GPT-4 [33]、LLaVA [25]、MiniGPT-4 [52]）进行了定性对比。尽管所有模型均能生成长篇回答，InstructBLIP 的输出通常包含更准确的视觉细节，并展示出逻辑连贯的推理步骤。值得注意的是，长篇回答并不总是最优选择。例如，在附录图 2 中，InstructBLIP 能够通过自适应调整回答长度直接响应用户意图，而 LLaVA 和 MiniGPT-4 则生成了冗长且相关性较低的句子。InstructBLIP 的这些优势源于多样化的指令微调数据和有效的架构设计。

3.4 指令微调 vs. 多任务学习

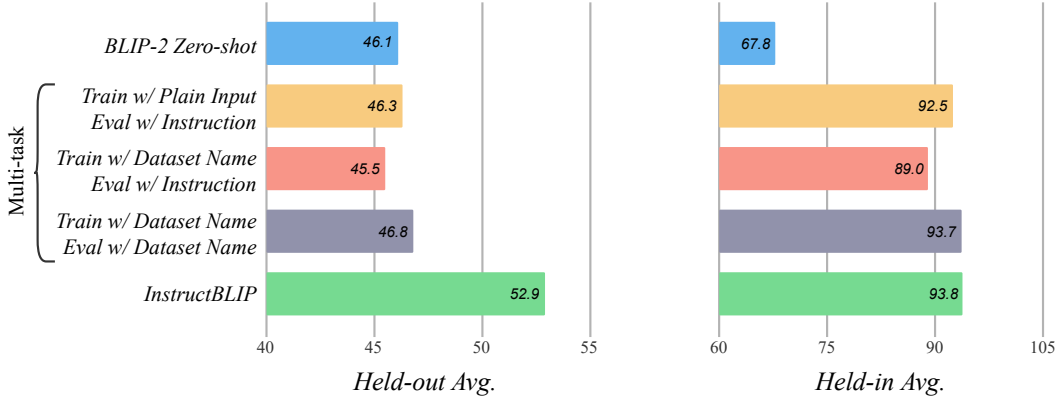


Figure 4: 基于 BLIP-2 FlanT5_{XL} 主干网络的指令微调与多任务训练对比。held-in 评估取所有 held-in 数据集的平均分；held-out 评估取 GQA、TextVQA、VSR、HatefulMemes、IconQA、ScienceQA、iVQA、VizWiz 的平均分。

与指令微调最为直接对应的方法是多任务学习——一种通过同时在多个数据集上训练以提升各数据集性能的常用方法。为探究指令微调在零样本泛化上的提升主要来自指令格式本身还是仅来自多任务训练，我们在相同训练设置下对两种方法进行了对比分析。

参照 [46]，我们考虑两种多任务训练方式：第一种直接使用各训练数据集的原始输入-输出格式，不含指令，但在评估时仍向模型提供指令以指明任务类型（图像描述任务除外，该任务仅输入图像时得分更高）。第二种在训练时于文本输入前添加 [Task:Dataset] 标识符（例如 VQAv2 数据集添加 [Visual question answering:VQAv2]），向指令微调迈进一步。评估时同时探索使用指令和该标识符两种方式；对于 held-out 数据集，由于模型从未见过数据集名称，仅使用任务名称。

图 4 展示了 BLIP-2 零样本、多任务训练和指令微调的结果，所有模型均基于 BLIP-2 FlanT5_{XL} 主干网络并采用第 2 节相同的训练配置。总体来看，可以得出两点结论：其一，指令微调与多任务学习在 held-in 数据集上表现相近，表明只要经过相应数据训练，模型对这两种输入模式的拟合能力相当。其二，在未见过的 held-out 数据集上，指令微调相比多任务学习有显著提升，而多任务学习仅与原始 BLIP-2 持平，说明指令微调是提升模型零样本泛化能力的关键。

3.5 在下游任务上微调 InstructBLIP

我们进一步对 InstructBLIP 模型进行微调，以研究其在特定数据集上的学习性能。与大多数先前方法（如 Flamingo、BLIP-2）在下游任务微调时提高输入图像分辨率并解冻视觉编码器不同，InstructBLIP 在指令微调阶段保持图像分辨率不变（224×224），在下游微调时也保持视觉编码器冻结。这使得可训练参数量从 1.2B 大幅降至 188M，显著提升了微调效率。

微调结果如表 3 所示。InstructBLIP 在所有数据集上的微调性能均优于 BLIP-2，验证了其作为任务特定微调更优初始化权重有效性。InstructBLIP 在 ScienceQA (IMG)、OCR-VQA、A-OKVQA 上均达到新的最优微调性能，仅在 OKVQA 上被参数量高达 562B 的 PaLM-E [9] 超越。

此外，我们发现基于 FlanT5 的 InstructBLIP 在多选任务上表现更优，而基于 Vicuna 的 InstructBLIP 在开放式生成任务上通常更胜一筹。这一差异主要源于两者冻结 LLM 的能力差异，因为二者使用相同的图像编码器。尽管 FlanT5 和 Vicuna 均为指令微调的 LLM，其训练数据存在显著差异：FlanT5 主要在包含大量多选问答和分类数据集的 NLP 基准上微调，而 Vicuna 则在开放式指令跟随数据上微调。

	ScienceQA IMG	OCR-VQA	OKVQA	A-OKVQA			
				Direct Val	Answer Test	Multi-choice Val	Test
Previous SOTA	LLaVA [25] 89.0	GIT [43] 70.3	PaLM-E(562B) [9] 66.1	[15] 56.3	[37] 61.6	[15] 73.2	[37] 73.6
BLIP-2 (FlanT5 _{XXL})	89.5	72.7	54.7	57.6	53.7	80.2	76.2
InstructBLIP (FlanT5 _{XXL})	90.7	73.3	55.5	57.1	54.8	81.0	76.7
BLIP-2 (Vicuna-7B)	77.3	69.1	59.3	60.0	58.7	72.1	69.0
InstructBLIP (Vicuna-7B)	79.5	72.8	62.1	64.0	62.1	75.7	73.4

Table 3: BLIP-2 与 InstructBLIP 在下游数据集上的微调结果。相比 BLIP-2, InstructBLIP 提供了更优的权重初始化, 并在四个数据集上的三个上达到最优性能。

4 相关工作

指令微调旨在训练语言模型遵循自然语言指令, 已被证明能够提升模型对未见任务的泛化性能。部分方法通过模板将现有 NLP 数据集转换为指令格式来收集指令微调数据 [46, 7, 35, 45]; 另一些方法则利用 LLM (如 GPT-3 [5]) 生成多样性更强的指令数据 [2, 13, 44, 40]。

经指令微调的 LLM 已通过注入视觉信息被适配用于视觉到语言的生成任务。BLIP-2 [20] 使用冻结的 FlanT5 模型, 并训练 Q-Former 将视觉特征作为输入提供给 LLM。MiniGPT-4 [52] 沿用 BLIP-2 的预训练视觉编码器和 Q-Former, 但将 LLM 替换为 Vicuna [2], 并使用 ChatGPT [1] 生成的比 BLIP-2 训练数据更长的图像描述进行训练。LLaVA [25] 直接将视觉编码器的输出投影为 LLaMA/Vicuna LLM 的输入, 并在 GPT-4 [33] 生成的视觉-语言对话数据上对 LLM 进行微调。mPLUG-owl [50] 使用文本指令数据和 LLaVA 的视觉-语言指令数据, 对 LLaMA [41] 模型进行低秩适配 [14] 微调。MultiInstruct [48] 则在无预训练 LLM 的情况下进行视觉-语言指令微调, 性能相对较弱。

与现有方法相比, InstructBLIP 使用了覆盖范围更广的视觉-语言指令数据, 同时涵盖基于模板转换的数据和 LLM 生成的数据。在架构层面, InstructBLIP 提出了指令感知的视觉特征提取机制。此外, 本文对视觉-语言指令微调的多个方面进行了全面分析, 验证了其在泛化到未见任务上的优势。

5 结论

本文提出 InstructBLIP —— 一个简洁而新颖的面向通用视觉-语言模型的指令微调框架。我们对视觉-语言指令微调进行了全面研究, 证明 InstructBLIP 模型能够以最优性能泛化到多种未见任务上。定性示例同样展示了 InstructBLIP 在指令遵循方面的多种能力, 包括复杂视觉推理、基于知识的图像描述和多轮对话。此外, 我们还证明 InstructBLIP 可作为下游任务微调的更优初始化权重, 并达到最优结果。我们期望 InstructBLIP 能够推动通用多模态 AI 及其应用领域的新研究。

References

- [1] Chatgpt. <https://openai.com/blog/chatgpt>, 2023. 8
- [2] Vicuna. <https://github.com/lm-sys/FastChat>, 2023. 3, 5, 8
- [3] Harsh Agrawal, Karan Desai, Yufei Wang, Xinlei Chen, Rishabh Jain, Mark Johnson, Dhruv Batra, Devi Parikh, Stefan Lee, and Peter Anderson. nocaps: novel object captioning at scale. In *ICCV*, pages 8948–8957, 2019. 3, 16
- [4] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob L Menick, Sebastian Borgeaud, Andy Brock, Aida Nematzadeh, Sahand Sharifzadeh, Miłoj aj Bińkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karén Simonyan. Flamingo: a visual language model for few-shot learning. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *NeurIPS*, 2022. 1, 6

- [5] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020. 8
- [6] Jaemin Cho, Jie Lei, Hao Tan, and Mohit Bansal. Unifying vision-and-language tasks via text generation. *arXiv preprint arXiv:2102.02779*, 2021. 1
- [7] Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Y. Zhao, Yanping Huang, Andrew M. Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*, 2022. 1, 3, 5, 8
- [8] Abhishek Das, Satwik Kottur, Khushi Gupta, Avi Singh, Deshraj Yadav, Jose M. F. Moura, Devi Parikh, and Dhruv Batra. Visual dialog. In *CVPR*, 2017. 3, 16
- [9] Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, and Pete Florence. Palm-e: An embodied multimodal language model, 2023. 7, 8
- [10] Yuxin Fang, Wen Wang, Binhui Xie, Quan-Sen Sun, Ledell Yu Wu, Xinggang Wang, Tiejun Huang, Xinlong Wang, and Yue Cao. Eva: Exploring the limits of masked visual representation learning at scale. *ArXiv*, abs/2211.07636, 2022. 5
- [11] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *CVPR*, July 2017. 3, 16
- [12] Danna Gurari, Qing Li, Abigale J. Stangl, Anhong Guo, Chi Lin, Kristen Grauman, Jiebo Luo, and Jeffrey P. Bigham. Vizwiz grand challenge: Answering visual questions from blind people. In *CVPR*, 2018. 3, 16
- [13] Or Honovich, Thomas Scialom, Omer Levy, and Timo Schick. Unnatural instructions: Tuning language models with (almost) no human labor. *ArXiv*, abs/2212.09689, 2022. 8
- [14] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022. 8
- [15] Yushi Hu, Hang Hua, Zhengyuan Yang, Weijia Shi, Noah A. Smith, and Jiebo Luo. Promptcap: Prompt-guided task-aware image captioning, 2023. 8
- [16] Drew A. Hudson and Christopher D. Manning. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *CVPR*, 2019. 3, 16
- [17] Andrej Karpathy and Li Fei-Fei. Deep visual-semantic alignments for generating image descriptions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2015. 16
- [18] Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. The hateful memes challenge: Detecting hate speech in multimodal memes. In *NeurIPS*, 2020. 3, 16
- [19] Dongxu Li, Junnan Li, Hung Le, Guangsen Wang, Silvio Savarese, and Steven C. H. Hoi. Lavis: A library for language-vision intelligence, 2022. 5
- [20] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *ICML*, 2023. 1, 4, 6, 8, 16
- [21] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *ICML*, 2022. 5, 16
- [22] Junnan Li, Ramprasaath Selvaraju, Akhilesh Gotmare, Shafiq Joty, Caiming Xiong, and Steven Chu Hong Hoi. Align before fuse: Vision and language representation learning with momentum distillation. In *NeurIPS*, 2021. 5

- [23] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco: Common objects in context. In *ECCV*, 2014. 3, 16
- [24] Fangyu Liu, Guy Edward Toh Emerson, and Nigel Collier. Visual spatial reasoning. *Transactions of the Association for Computational Linguistics*, 2023. 3
- [25] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. 2023. 3, 6, 8, 16
- [26] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *ICLR*, 2019. 5
- [27] Jiasen Lu, Vedanuj Goswami, Marcus Rohrbach, Devi Parikh, and Stefan Lee. 12-in-1: Multi-task vision and language representation learning. In *CVPR*, 2020. 1
- [28] Pan Lu, Swaroop Mishra, Tony Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. Learn to explain: Multimodal reasoning via thought chains for science question answering. In *NeurIPS*, 2022. 3, 16
- [29] Pan Lu, Liang Qiu, Jiaqi Chen, Tony Xia, Yizhou Zhao, Wei Zhang, Zhou Yu, Xiaodan Liang, and Song-Chun Zhu. Iconqa: A new benchmark for abstract diagram understanding and visual language reasoning. In *NeurIPS Track on Datasets and Benchmarks*, 2021. 3, 16
- [30] Kenneth Marino, Mohammad Rastegari, Ali Farhadi, and Roozbeh Mottaghi. Ok-vqa: A visual question answering benchmark requiring external knowledge. In *CVPR*, 2019. 3, 16
- [31] Anand Mishra, Shashank Shekhar, Ajeet Kumar Singh, and Anirban Chakraborty. Ocr-vqa: Visual question answering by reading text in images. In *ICDAR*, 2019. 3, 16
- [32] Vishvak Murahari, Dhruv Batra, Devi Parikh, and Abhishek Das. Large-scale pretraining for visual dialog: A simple state-of-the-art baseline. In Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm, editors, *ECCV*, 2020. 5, 6
- [33] OpenAI. Gpt-4 technical report. *ArXiv*, abs/2303.08774, 2023. 6, 8
- [34] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *The Journal of Machine Learning Research*, 2020. 3, 5
- [35] Victor Sanh, Albert Webson, Colin Raffel, Stephen H. Bach, Lintang Sutawika, Zaid Alyafeai, Antoine Chaffin, Arnaud Stiegler, Arun Raja, Manan Dey, M Saiful Bari, Canwen Xu, Urmish Thakker, Shanya Sharma Sharma, Eliza Szczechla, Taewoon Kim, Gunjan Chhablani, Nihal V. Nayak, Debajyoti Datta, Jonathan Chang, Mike Tian-Jian Jiang, Han Wang, Matteo Manica, Sheng Shen, Zheng Xin Yong, Harshit Pandey, Rachel Bawden, Thomas Wang, Trishala Neeraj, Jos Rozen, Abheesht Sharma, Andrea Santilli, Thibault Févry, Jason Alan Fries, Ryan Teehan, Teven Le Scao, Stella Biderman, Leo Gao, Thomas Wolf, and Alexander M. Rush. Multitask prompted training enables zero-shot task generalization. In *ICLR*, 2022. 8
- [36] Dustin Schwenk, Apoorv Khandelwal, Christopher Clark, Kenneth Marino, and Roozbeh Mottaghi. A-okvqa: A benchmark for visual question answering using world knowledge. In Shai Avidan, Gabriel Brostow, Moustapha Cissé, Giovanni Maria Farinella, and Tal Hassner, editors, *ECCV*, 2022. 3, 16
- [37] Zhenwei Shao, Zhou Yu, Meng Wang, and Jun Yu. Prompting large language models with answer heuristics for knowledge-based visual question answering. *Computer Vision and Pattern Recognition (CVPR)*, 2023. 8
- [38] Oleksii Sidorov, Ronghang Hu, Marcus Rohrbach, and Amanpreet Singh. Textcaps: a dataset for image captioning with reading comprehension. 2020. 3, 16
- [39] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read. In *CVPR*, pages 8317–8326, 2019. 3, 16
- [40] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca, 2023. 8
- [41] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023. 3, 5, 8
- [42] Ramakrishna Vedantam, C. Lawrence Zitnick, and Devi Parikh. Cider: Consensus-based image description evaluation. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4566–4575, 2015. 6

- [43] Jianfeng Wang, Zhengyuan Yang, Xiaowei Hu, Linjie Li, Kevin Lin, Zhe Gan, Zicheng Liu, Ce Liu, and Lijuan Wang. Git: A generative image-to-text transformer for vision and language, 2022. 8
- [44] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language model with self generated instructions. *ArXiv*, abs/2212.10560, 2022. 8
- [45] Yizhong Wang, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Atharva Naik, Arjun Ashok, Arut Selvan Dhanasekaran, Anjana Arunkumar, David Stap, Eshaan Pathak, Giannis Karamanolakis, Haizhi Lai, Ishan Purohit, Ishani Mondal, Jacob Anderson, Kirby Kuznia, Krima Doshi, Kuntal Kumar Pal, Maitreya Patel, Mehrad Moradshahi, Mihir Parmar, Mirali Purohit, Neeraj Varshney, Phani Rohitha Kaza, Pulkit Verma, Ravsehaj Singh Puri, Rushang Karia, Savan Doshi, Shailaja Keyur Sampat, Siddhartha Mishra, Sujan Reddy A, Sumanta Patro, Tanay Dixit, and Xudong Shen. Super-NaturalInstructions: Generalization via declarative instructions on 1600+ NLP tasks. In *EMNLP*, 2022. 8
- [46] Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. Finetuned language models are zero-shot learners. In *ICLR*, 2022. 1, 5, 7, 8
- [47] Dejing Xu, Zhou Zhao, Jun Xiao, Fei Wu, Hanwang Zhang, Xiangnan He, and Yueting Zhuang. Video question answering via gradually refined attention over appearance and motion. In *Proceedings of the 25th ACM International Conference on Multimedia*, page 1645 – 1653, 2017. 3, 16
- [48] Zhiyang Xu, Ying Shen, and Lifu Huang. Multiinstruct: Improving multi-modal zero-shot learning via instruction tuning. *ArXiv*, abs/2212.10773, 2022. 8
- [49] Antoine Yang, Antoine Miech, Josef Sivic, Ivan Laptev, and Cordelia Schmid. Just ask: Learning to answer questions from millions of narrated videos. In *ICCV*, pages 1686–1697, 2021. 3, 6, 16
- [50] Qinghao Ye, Haiyang Xu, Guohai Xu, Jiabo Ye, Ming Yan, Yi Zhou, Junyan Wang, Anwen Hu, Pengcheng Shi, Yaya Shi, Chenliang Li, Yuanhong Xu, Hehong Chen, Junfeng Tian, Qiang Qi, Ji Chao Zhang, and Feiyan Huang. mplug-owl: Modularization empowers large language models with multimodality. 2023. 8
- [51] Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Transactions of the Association for Computational Linguistics*, 2, 2014. 3, 16
- [52] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models, 2023. 6, 8

A Broader Impact

InstructBLIP uses off-the-shelf frozen LLMs. Therefore it inherits some of the shortcomings from the original LLMs, such as hallucinating ungrounded text or generating outputs with bias. We mitigate such shortcomings by improving the model’s grounding on the vision and instruction input, and performing vision-language instruction tuning on a diverse set of high-quality datasets. Nevertheless, we do not recommend applying InstructBLIP models to any downstream applications without a prior assessment on safety and fairness specific to that application.

B More Case Studies

[Input Image]	
[Input Instruction]	What is unusual about this image?
[Model Response]	
InstructBLIP	 The unusual aspect of this image is that a man is ironing clothes on the back of a yellow taxi cab while it is driving down a bus city street. He is standing on top of the car's trunk, holding an iron and using it to press his clothing items. This situation is not only unconventional but also potentially hazardous, as the man could be at risk of falling off the vehicle or causing an accident due to his actions.
GPT-4	 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.
MiniGPT-4	The image shows a man standing on the back of a yellow taxi, with a yellow shirt and black pants, and a blue backpack on his back. The taxi is driving on a city street with other cars and taxis in the background.

Figure 5: The response from InstructBLIP is more comprehensive than GPT-4, more visually-grounded than LLaVA, and more logical than MiniGPT-4. The responses of GPT-4 and LLaVA are obtained from their respective papers, while the official demo is used for MiniGPT-4. Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

[Input Image]	
	

[Input Instruction] Do you know who drew this painting?	

[Model Response]	
InstructBLIP 	Leonardo da Vinci.
LLaVA	The painting depicts a woman, commonly believed to be Mona Lisa, the famous artwork by Leonardo da Vinci. It is a portrait painting that showcases the woman's enigmatic smile and has become one of the most famous and iconic art pieces in the world. The original work is displayed in the Louvre Museum in Paris, and it is known for its intricate details, use of oil paint, and the artist's innovative techniques that contributed to its enduring appeal and mystery.
MiniGPT-4	The painting is the Mona Lisa, also known as the Mona Lisa, by Leonardo da Vinci. It is one of the most famous paintings in the world and is known for its enigmatic smile and the mysterious identity of the sitter. The painting is housed in the Louvre Museum in Paris, France.

Figure 6: In this example, we illustrate that long-form responses are not always preferable. InstructBLIP can directly address the user’s intent by adaptively adjusting the response length, while other models tend to generate lengthy paragraphs with less-relevant sentences. The response from LLaVA is taken from the paper, and for MiniGPT-4, we utilize its official demo.

[Input Image]	
[Input Instruction]	Introduce me this painting in detail.
[Model Response]	
InstructBLIP	 The painting depicts a young girl with a pearl earring, also known as "The Girl with the Pearl Earring" or "Girl with a Pearl Earring". She is wearing a blue dress and has a pearl earring on her left ear. Her hair is pulled back into a bun, and she is looking directly at the viewer. This painting is a famous work of art by Dutch painter Johannes Vermeer, created in 1665. It is considered one of his masterpieces and is widely recognized for its intricate details and realistic portrayal of the young girl's facial expression.
LLaVA	The painting features a woman with a headscarf, possibly of Dutch origin, dressed in blue and white clothing. She is wearing a large blue hat and has a red mouth, which adds a vibrant touch to the portrait. The woman is looking directly at the viewer, creating a sense of connection. The background of the painting is predominantly blue, further emphasizing the color theme of the piece. The overall composition and attire of the woman suggest that the painting might be inspired by the style of Rembrandt.
MiniGPT-4	The painting depicts a young woman with a pearl earring, wearing a blue dress and a white apron. She is shown with a serious expression, looking directly at the viewer. The background is a dark, neutral tone, with no visible objects or elements. The overall mood of the painting is somber and contemplative. The painting is well-executed, with a high level of detail and realism in the depiction of the woman's features and clothing. The use of shading and lighting creates a sense of depth and dimension, making the woman appear three-dimensional. The artist has used a limited color palette, with the woman's dress and apron providing the only sources of color in the painting. The painting is well-composed, with the woman's head and shoulders positioned in the center of the canvas, creating a sense of balance and symmetry. The use of negative space around the woman's figure adds to the overall somber mood of the painting. Overall, this is a well-executed and thought-provoking painting that effectively conveys a sense of contemplation and introspection.

Figure 7: InstructBLIP shows the ability to connect textual knowledge with visual input, while the other models depict the picture plainly. MiniGPT-4 exhibits poorer results, which may be due to its training with only long captions. Responses of LLaVA and MiniGPT-4 are generated by their official demos.

C Instruction Tuning Datasets

Dataset Name	Held-out	Dataset Description
COCO Caption [23]	✗	We use the large-scale COCO dataset for the image captioning task. Specifically, Karpathy split [17] is used, which divides the data into 82K/5K/5K images for the train/val/test sets.
Web CapFilt	✗	14M image-text pairs collected from the web with additional BLIP-generated synthetic captions, used in BLIP [21] and BLIP-2 [20].
NoCaps [3]	✓ (val)	NoCaps contains 15,100 images with 166,100 human-written captions for novel object image captioning.
Flickr30K [51]	✓ (test)	The Flickr30k dataset consists of 31K images collected from Flickr, each image has five ground truth captions. We use the test split as the held-out which contains 1K images.
TextCaps [38]	✗	TextCaps is an image captioning dataset that requires the model to comprehend and reason the text in images. Its train/val/test sets contain 21K/3K/3K images, respectively.
VQAv2 [11]	✗	VQAv2 is dataset for open-ended image question answering. It is split into 82K/40K/81K for train/val/test.
VizWiz [12]	✓ (test-dev)	A dataset contains visual questions asked by people who are blind. 8K images are used for the held-out evaluation.
GQA [16]	✓ (test-dev)	GQA contains image questions for scene understanding and reasoning. We use the balanced test-dev set as held-out.
Visual Spatial Reasoning	✓ (test)	VSR is a collection of image-text pairs, in which the text describes the spatial relation of two objects in the image. Models are required to classify true/false for the description. We use the zero-shot data split given in its official github repository.
IconQA [29]	✓ (test)	IconQA measures the abstract diagram understanding and comprehensive cognitive reasoning abilities of models. We use the test set of its multi-text-choice task for held-out evaluation.
OKVQA [30]	✗	OKVQA contains visual questions that require outside knowledge to answer. It has been split into 9K/5K for train and test.
A-OKVQA [36]	✗	A-OKVQA is a successor of OKVQA with more challenging and diverse questions. It has 17K/1K/6K questions for train/val/test.
ScienceQA [28]	✓ (test)	ScienceQA covers diverse science topics with corresponding lectures and explanations. In our settings, we only use the part with image context (IMG).
Visual Dialog [8]	✓ (val)	Visual dialog is a conversational question answering dataset. We use the val split as the held-out, which contains 2,064 images and each has 10 rounds.
OCR-VQA [31]	✗	OCR-VQA contains visual questions that require models to read text in the image. It has 800K/100K/100K for train/val/test, respectively.
TextVQA [39]	✓ (val)	TextVQA requires models to comprehend visual text to answer questions.
HatefulMemes [18]	✓ (val)	A binary classification dataset to justify whether a meme contains hateful content.
LLaVA-Instruct-150K [25]	✗	An instruction tuning dataset which has three parts: detailed caption (23K), reasoning (77K), conversation (58K).
MSVD-QA [47]	✓ (test)	We use the test set (13K video QA pairs) of MSVD-QA for held-out testing.
MSRVTT-QA [47]	✓ (test)	MSRVTT-QA has more complex scenes than MSVD, with 72K video QA pairs as the test set.
iVQA [49]	✓ (test)	iVQA is a video QA dataset with mitigated language biases. It has 6K/2K/2K samples for train/val/test.

Table 4: Description of datasets in our held-in instruction tuning and held-out zero-shot evaluations.

D Instruction Templates

Task	Instruction Template
Image Captioning	<Image>A short image caption: <Image>A short image description: <Image>A photo of <Image>An image that shows <Image>Write a short description for the image. <Image>Write a description for the photo. <Image>Provide a description of what is presented in the photo. <Image>Briefly describe the content of the image. <Image>Can you briefly explain what you see in the image? <Image>Could you use a few words to describe what you perceive in the photo? <Image>Please provide a short depiction of the picture. <Image>Using language, provide a short account of the image. <Image>Use a few words to illustrate what is happening in the picture.
VQA	<Image>{Question} <Image>Question: {Question} <Image>{Question} A short answer to the question is <Image>Q: {Question} A: <Image>Question: {Question} Short answer: <Image>Given the image, answer the following question with no more than three words. {Question} <Image>Based on the image, respond to this question with a short answer: {Question}. Answer: <Image>Use the provided image to answer the question: {Question} Provide your answer as short as possible: <Image>What is the answer to the following question? "{Question}" <Image>The question "{Question}" can be answered using the image. A short answer is
VQG	<Image>Given the image, generate a question whose answer is: {Answer}. Question: <Image>Based on the image, provide a question with the answer: {Answer}. Question: <Image>Given the visual representation, create a question for which the answer is "{Answer}". <Image>From the image provided, craft a question that leads to the reply: {Answer}. Question: <Image>Considering the picture, come up with a question where the answer is: {Answer}. <Image>Taking the image into account, generate an question that has the answer: {Answer}. Question:

Table 5: Instruction templates used for transforming held-in datasets into instruction tuning data. For datasets with OCR tokens, we simply add “OCR tokens:” after the image query embeddings.

E Instructions for Zero-shot Inference

We provide instructions used for zero-shot inference. Note that for instructions with options, we separate options with the alphabetical order, e.g. (a) blue (b) yellow (c) pink (d) black.

GQA, VizWiz, iVQA, MSVD, MSRVT <Image> Question: {} Short answer:

NoCaps, Flickr30k <Image> A short image description:

TextVQA <Image> OCR tokens: {}. Question: {} Short answer:

IconQA <Image> Question: {} Options: {}. Short answer:

ScienceQA <Image> Context: {} Question: {} Options: {}. Answer:

HatefulMemes <Image> This is an image with: "{}" written on it. Is it hateful? Answer:

VSR <Image> Based on the image, is this statement true or false? "{}" Answer:

Visual Dialog <Image> Dialog history: {} \n Question: {} Short answer: